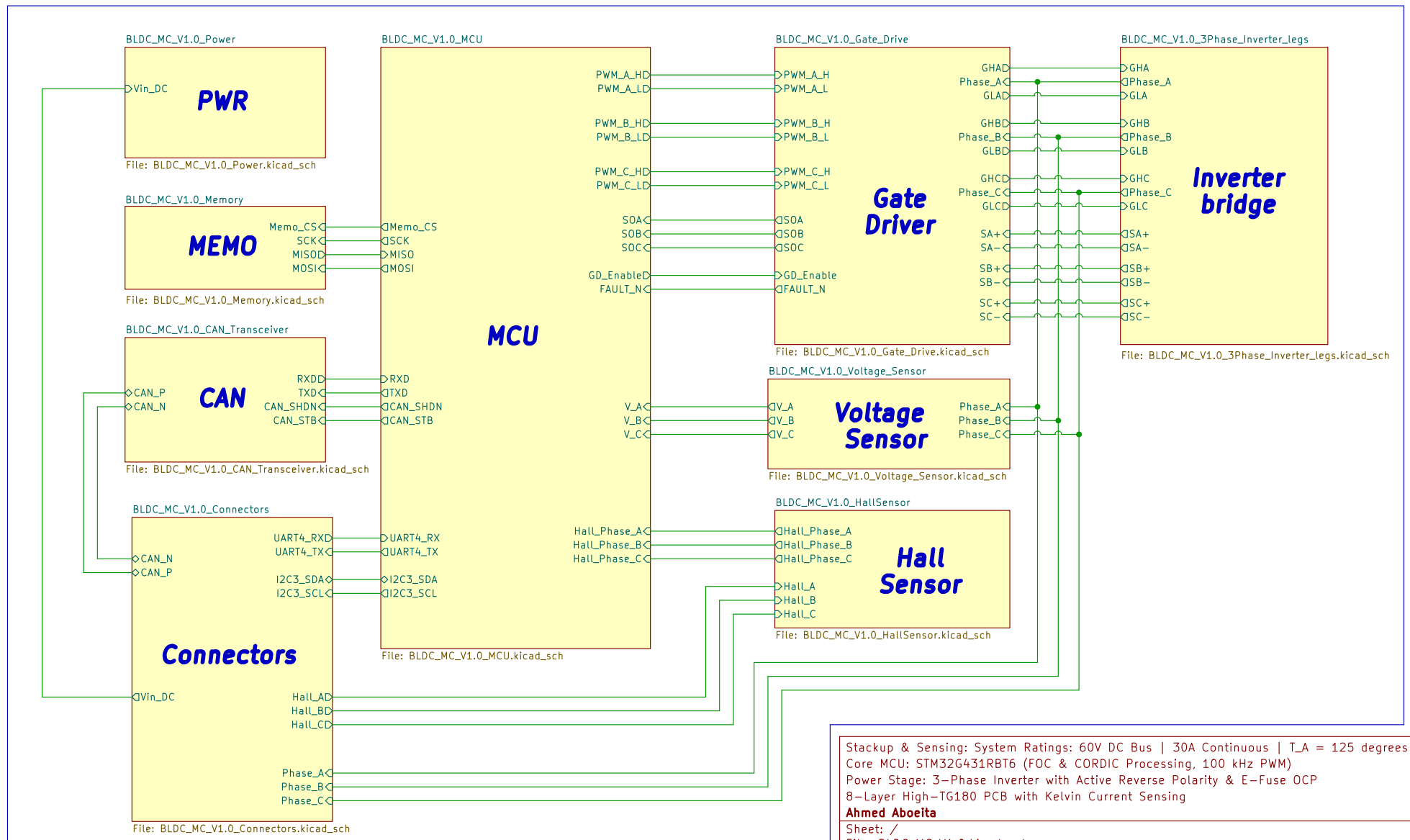


[1] BLDC_MC V1 60V 30A



Stackup & Sensing: System Ratings: 60V DC Bus | 30A Continuous | T_A = 125 degrees
Core MCU: STM32G431RBT6 (FOC & CORDIC Processing, 100 kHz PWM)
Power Stage: 3-Phase Inverter with Active Reverse Polarity & E-Fuse OCP
8-Layer High-TG180 PCB with Kelvin Current Sensing

Ahmed Aboeita

Sheet: /
File: BLDC_MC_V1.0.kicad_sch

Title: BLDC_MC

Size: A4	Date: 2026-08-17
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KiCad E.D.A. 10.0.5

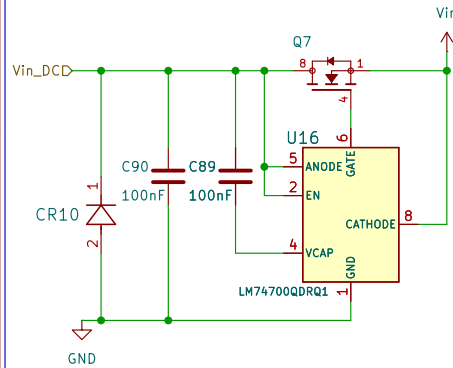
Rev: V1.0

Id: 1/10

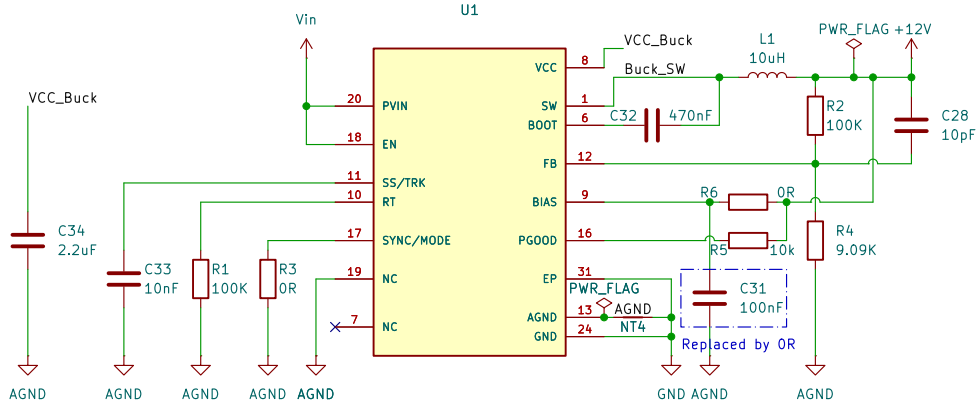
[1] Power Conversion Vin to 12V and 5V to 3.3V

The maximum input voltage of this 60V so we can use for BLDC motors up to 60V

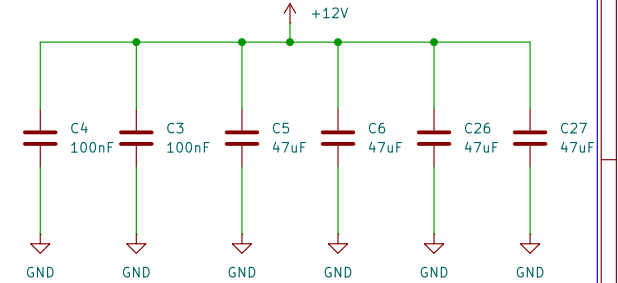
Input PWR Protection



Buck IC Circuit

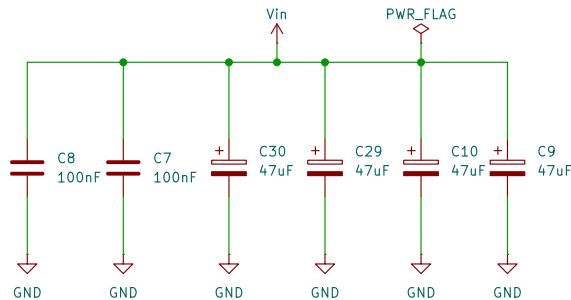


Output Cap Bank



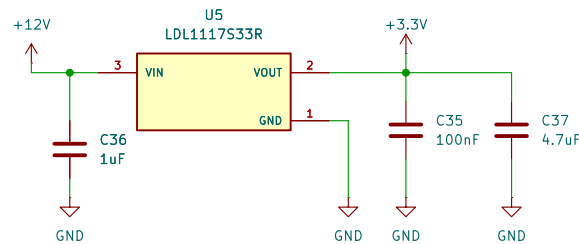
Output Capacitor Bank
Recommended from IC datasheet
Cout = 150uF
Reference design puts 5 of 47uF

Input Cap Bank

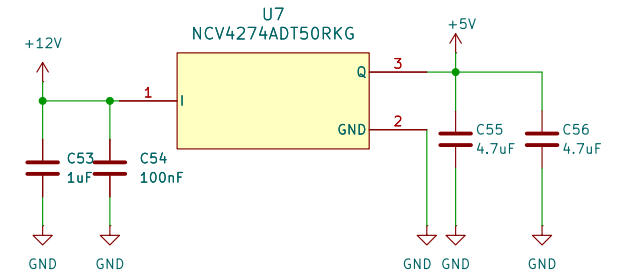


input Capacitor Bank
To make the input power pure DC

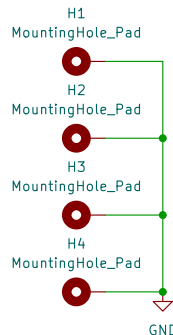
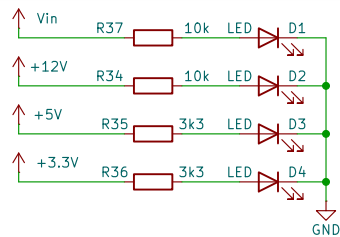
LDO Fixed out 3.3V



LDO Fixed out 5V



Indication Leds



Voltage Level	Usage
12V	Gate drive and MOSFET firing Voltage
5V	for Hall Sensor
3.3V	Controller and aux ICs

Stackup & Sensing: System Ratings: 60V DC Bus | 30A Continuous | T_A = 125 degrees
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Power Stage: 3-Phase Inverter with Active Reverse Polarity & E-Fuse OCP
8-Layer High-TG180 PCB with Kelvin Current Sensing

Ahmed Aboeita

Sheet: /BLDC_MC_V1.0_Power/
File: BLDC_MC_V1.0_Power.kicad_sch

Title: BLDC_MC

Size: A4 Date: 2026-08-17

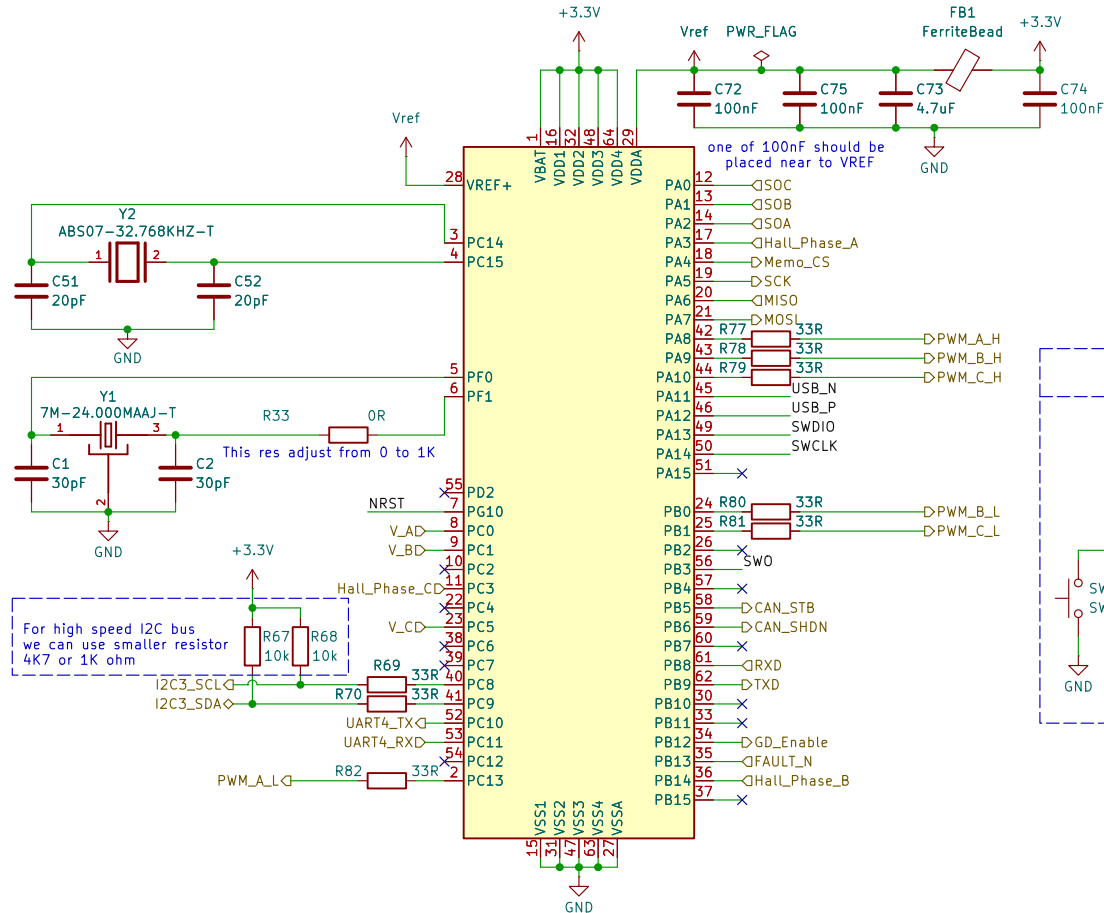
KiCad E.D.A. 10.0.5

Rev: V1.0

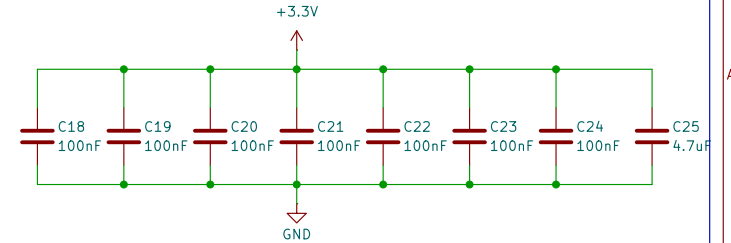
Id: 2/10

[3] STM32G431RBT6 Controller Interface

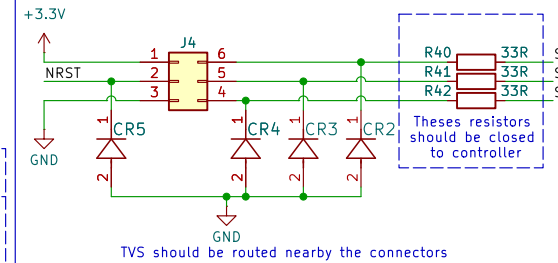
This MCU can handle the FOC control Algorithm



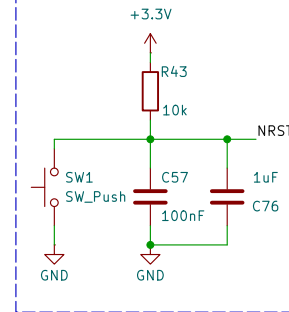
Power bypass Capacitors



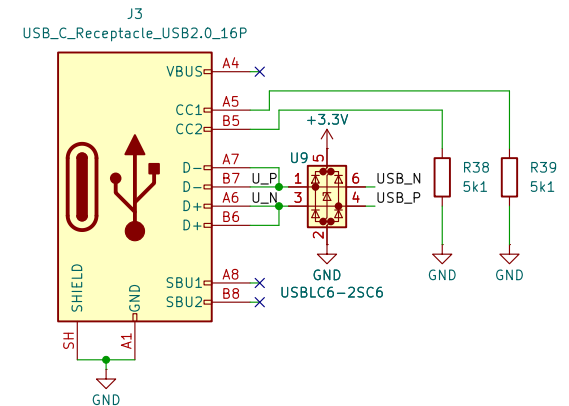
Debugger Interface (Serial Wire Debugger)



NRST circuit



USB-C



Stackup & Sensing: System Ratings: 60V DC Bus | 30A Continuous | T_A = 125 degrees
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Power Stage: 3-Phase Inverter with Active Reverse Polarity & E-Fuse OCP
8-Layer High-TG180 PCB with Kelvin Current Sensing

Ahmed Aboeita

Sheet: /BLDC_MC_V1.0_MCU/
File: BLDC_MC_V1.0_MCU.kicad_sch

Title: BLDC_MC

Size: A4 Date: 2026-08-17
KiCad E.D.A. 10.0.5

Rev: V1.0
Id: 3/10

for the 3 phase inverter with current sensor

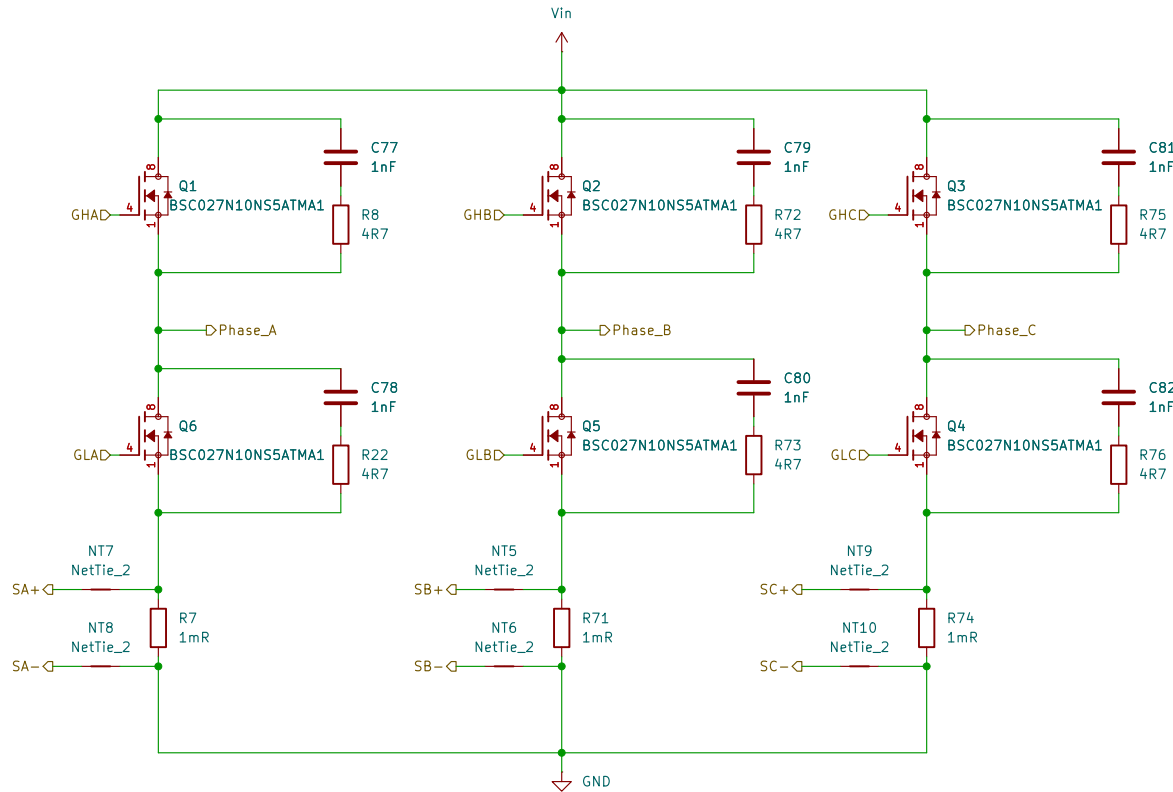


Smart Driver: Uses an internal, programmable current source called IDRIVE. You set the source/sink current (e.g., 100mA, 1A) via a single pin or SPI. This allows you to tune the switching speed of your inverter without changing any hardware components.

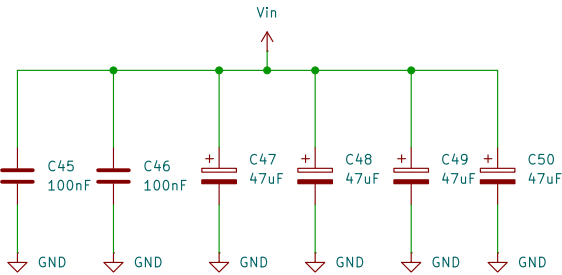
Rev: V1.0
Id: 4/10

[5] 3 Phases Inverter Bridge

3 Phase Inverter Bridge



DC Link



Steps for adjusting snubber circuit

- 1- Measure the Initial Ringing
- 2- Add the "Shift" Capacitor - increase the value of C_add until the ringing frequency is exactly half of the original frequency $f_{new} = 0.5 * f_{ring}$.
- 3- Calculate the parasitic capacitance $C_p = C_{add} / 3$
- 4- Calculate Parasitic Inductance $L_p \rightarrow L_p = 1 / ((2 * \pi * f_{ring})^2 * C_p)$
- 5- Determine the Snubber Resistor $R_{snub} \rightarrow R_{snub} = \sqrt{L_p / C_p}$

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Core MCU: STM32G431RBT6 (FOC & CORDIC Processing, 100 kHz PWM)
Power Stage: 3-Phase Inverter with Active Reverse Polarity & E-Fuse OCP
8-Layer High-TG180 PCB with Kelvin Current Sensing

Ahmed Aboeita

Sheet: /BLDC_MC_V1.0_3Phase_Inverter_legs/
File: BLDC_MC_V1.0_3Phase_Inverter_legs.kicad_sch

Title: BLDC_MC

Size: A4

Date: 2026-08-17

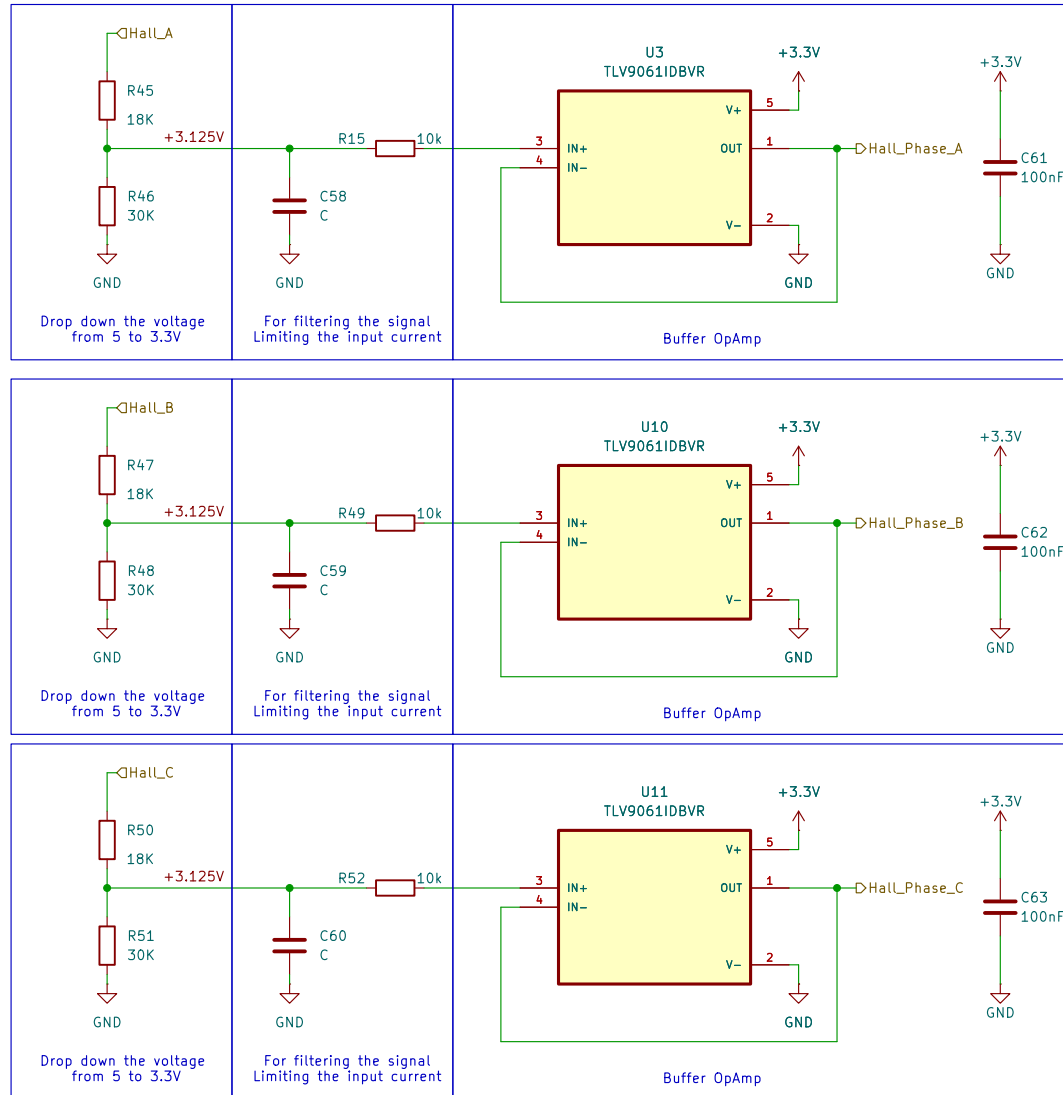
Rev: V1.0

KiCad E.D.A. 10.0.5

Id: 5/10

[6] Hall Effect Sensor Interface

For the 3 Phases of Hall Sensor



Stackup & Sensing: System Ratings: 60V DC Bus | 30A Continuous | T_A = 125 degrees
Core MCU: STM32G431RBT6 (FOC & CORDIC Processing, 100 kHz PWM)
Power Stage: 3-Phase Inverter with Active Reverse Polarity & E-Fuse OCP
8-Layer High-TG180 PCB with Kelvin Current Sensing

Ahmed Aboeita

Sheet: /BLDC_MC_V1.0_HallSensor/
File: BLDC_MC_V1.0_HallSensor.kicad_sch

Title: BLDC_MC

Size: A4

Date: 2026-08-17

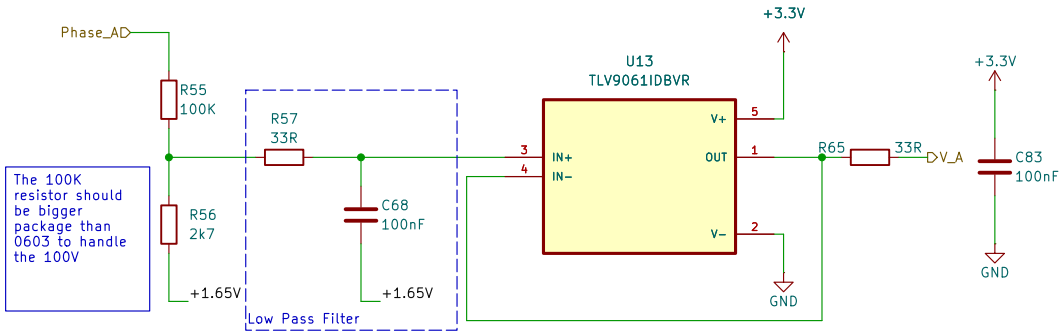
Rev: V1.0

KiCad E.D.A. 10.0.5

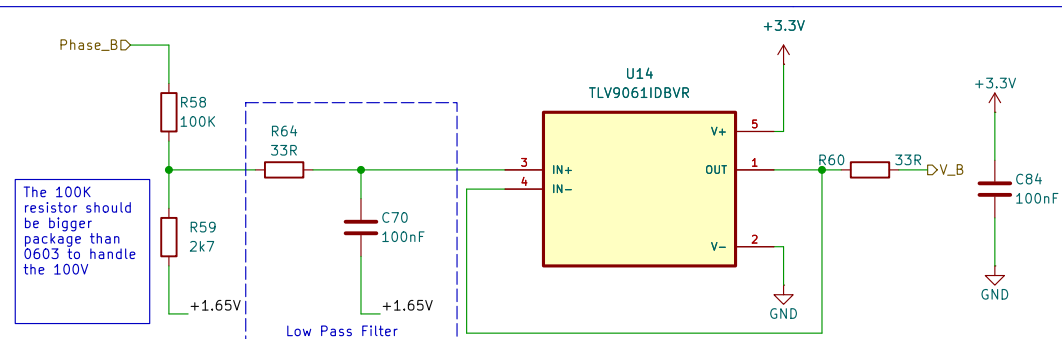
Id: 6/10

[7] 3 Phase Inverter Voltage Sensing

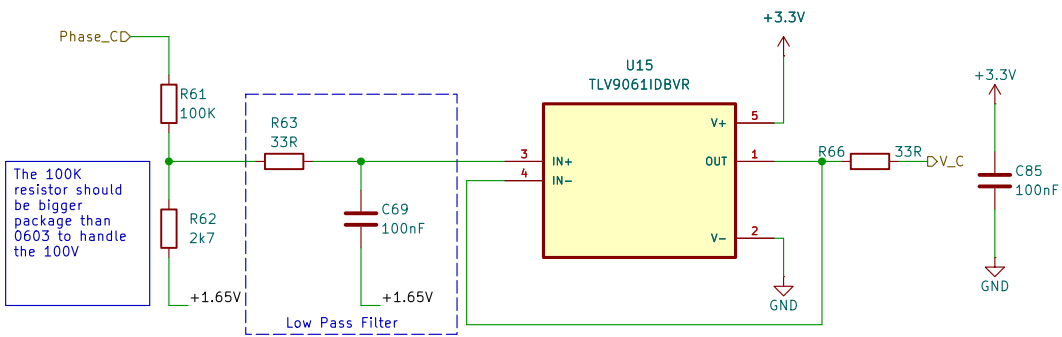
Phase A



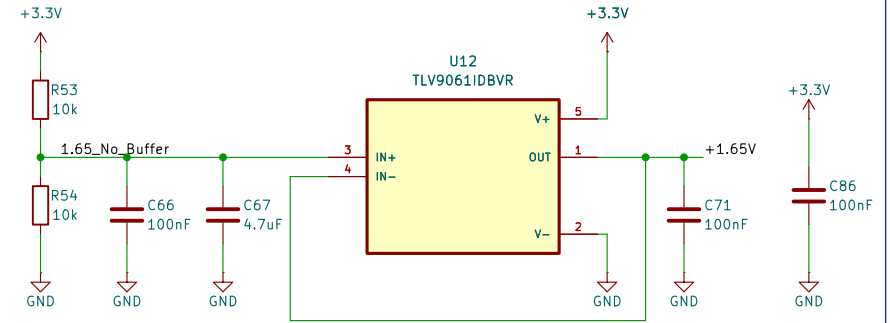
Phase B



Phase C



Virtual Ground +1.65V



Stackup & Sensing: System Ratings: 60V DC Bus | 30A Continuous | T_A = 125 degrees
Core MCU: STM32G431RBT6 (FOC & CORDIC Processing, 100 kHz PWM)
Power Stage: 3-Phase Inverter with Active Reverse Polarity & E-Fuse OCP
8-Layer High-TG180 PCB with Kelvin Current Sensing

Ahmed Aboeita

Sheet: /BLDC_MC_V1.0_Voltage_Sensor/
File: BLDC_MC_V1.0_Voltage_Sensor.kicad_sch

Title: BLDC_MC

Size: A4 Date: 2026-08-17

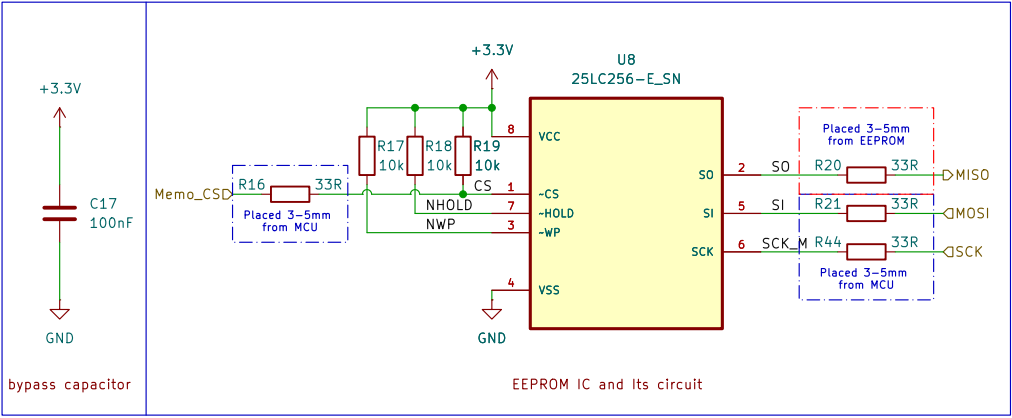
KiCad E.D.A. 10.0.5

Rev: V1.0

Id: 7/10

[8] EEPROM Memory 256Kbit

This memory used to store important data and configuration parameters



Stackup & Sensing: System Ratings: 60V DC Bus | 30A Continuous | T_A = 125 degrees
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Power Stage: 3-Phase Inverter with Active Reverse Polarity & E-Fuse OCP
8-Layer High-TG180 PCB with Kelvin Current Sensing

Ahmed Aboeita

Sheet: /BLDC_MC_V1.0_Memory/
File: BLDC_MC_V1.0_Memory.kicad_sch

Title: BLDC_MC

Size: A4 Date: 2026-08-17

KiCad E.D.A. 10.0.5

Rev: V1.0

Id: 8/10

This IC can operate as CAN-FD for faster data rate



Series Resistors Help manage signal integrity by damping potential high-frequency noise or reflections on the digital traces.

Stackup & Sensing: System Ratings: 60V DC Bus | 30A Continuous | T_A = 125 degrees
Core MCU: STM32G431RBT6 (FOC & CORDIC Processing, 100 kHz PWM)
Power Stage: 3-Phase Inverter with Active Reverse Polarity & E-Fuse OCP
8-Layer High-Tg180 PCB with Kelvin Current Sensing

Sheet: /BLDC_MC_V1.0_CAN_Transceiver/

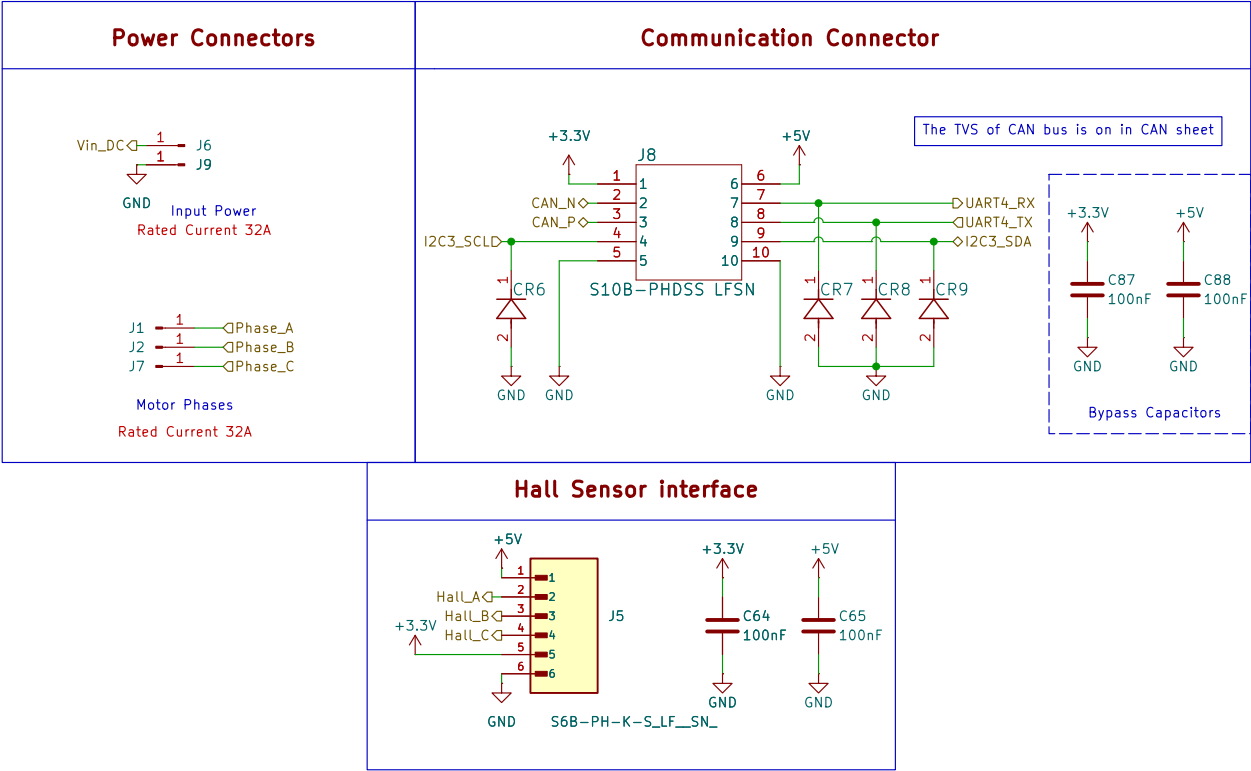
Title: BLDC_MC

Date: 2026-08-17

Rev: V1.0

Id: 9/10

[10] Board's Connectors Power & Communication



Stackup & Sensing: System Ratings: 60V DC Bus | 30A Continuous | T_A = 125 degrees
Core MCU: STM32G431RBT6 (FOC & CORDIC Processing, 100 kHz PWM)
Power Stage: 3-Phase Inverter with Active Reverse Polarity & E-Fuse OCP
8-Layer High-TG180 PCB with Kelvin Current Sensing

Ahmed Aboeita

Sheet: /BLDC_MC_V1.0_Connectors/
File: BLDC_MC_V1.0_Connectors.kicad_sch

Title: BLDC_MC

Size: A4 Date: 2026-08-17

KiCad E.D.A. 10.0.5

Rev: V1.0

Id: 10/10